

MILESTONE 1 · DS & AI LAB · MAY 2026

# Multimodal Damage Assessment for Insurance Claims

Problem Definition & Literature Review

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# The Problem

## Manual. Slow. Inconsistent.

When a vehicle is damaged, a claims assessor must:

- Manually review all submitted photographs
- Cross-reference the policyholder's insurance document
- Write a preliminary assessment report

This process takes one to several business days and is prone to inter-assessor variability.

**We build an AI decision-support system to automate the initial assessment stage.**

## Stakeholders

PRIMARY

### Claim Assessors

Faster first-pass reports; less repetitive manual work

PRIMARY

### Insurance Companies

Reduced processing time; standardised initial assessments

SECONDARY

### Policyholders

Faster decisions; transparent, traceable documentation

# Scope Definition

## In Scope

- Damage detection & localisation via fine-tuned YOLO
- Classification: dent, scratch, crack, broken lamp, flat tyre, shattered glass
- Severity estimation: Minor / Moderate / Severe per damage region
- Policy clause retrieval via RAG from uploaded PDF
- LLM-generated preliminary assessment report
- Gradio web interface on Hugging Face Spaces

## Out of Scope

- Final claim approval (human assessor decides)
- Repair cost estimation (requires make, model, labour rates)
- Internal / mechanical damage detection
- Real insurer policies (synthetic policies used)
- Multi-vehicle accidents or fraud detection
- Third-party liability assessment

# Why This Matters

**\$10B  
to \$15B**

India Motor Insurance Market  
2026 → 2031

## Throughput Bottleneck

Manual photo review is the largest time  
cost per claim

## Assessor Variability

Same damage, different severity ratings  
between assessors

## Policy Cross-referencing

Multi-page policy docs read repeatedly  
for every claim

## Scalability

Surge events (storms, floods) flood the  
pipeline

# Prior Research: Computer Vision

## Foundational

### Patil et al. (2017)

First CNN-based binary classifier for vehicle damage. Established that deep learning could distinguish damaged from undamaged vehicles.

## Detection

### YOLOv8 IEEE (2024)

Fine-tuned on 4,000+ images with 21 car-part classes & 8 damage types. Strong mAP for simultaneous part + damage segmentation.

## SOTA

### HL-YOLO MDPI (2025)

+2.5% precision, +5.8% recall, +3-4% mAP over YOLO11 baseline via heterogeneous convolutions. Closest to our chosen approach.

## Two-Stage

### Mask R-CNN He et al. (2017)

Higher segmentation fidelity via vehicle-then-damage pipeline. Achieves better masks but at significantly greater compute cost.

# Prior Research: RAG & Industry

## RAG for Document Understanding

### Chunk Size Matters

200-400 token chunks with overlap consistently outperform large-chunk retrieval for precise clause-level recall in legal documents.

### Bi-encoders Beat BM25

MiniLM-L6-v2 and MPNet outperform sparse BM25 retrieval for semantic matching of insurance-style queries.

### Faithfulness is Critical

Without grounding, LLMs hallucinate coverage entitlements. RAG with source attribution substantially reduces this (ES-RAG, 2024).

## Industry & Academic Landscape

### Industry (proprietary):

Tractable, CCC Intelligent Solutions, Mitchell

Deployed CV systems in production. However:

- Technical details are proprietary
- No policy grounding in any public demo
- Inaccessible to researchers / small insurers

### Closest Academic Analogue:

MIMIC-CXR: vision model detects, LLM generates a structured report. We adapt this paradigm to vehicle damage assessment.

# Literature Comparison Summary

## Structured Critical Comparison of Key Prior Works Reviewed

| Paper / System                          | Dataset                                      | Model / Method                      | Metrics (Scores)                             | Key Limitations                                             |
|-----------------------------------------|----------------------------------------------|-------------------------------------|----------------------------------------------|-------------------------------------------------------------|
| Patil et al. (2017)                     | Custom small dataset                         | CNN (classification only)           | ~84% binary accuracy on held-out test set    | No localisation; very small dataset; no severity estimation |
| He et al. Mask R-CNN (ICCV 2017)        | COCO                                         | Mask R-CNN                          | Box AP: 37.1; Mask AP: 35.7 (COCO test-dev)  | High compute cost; requires pre-segmented vehicle           |
| CarDD benchmark (USTC, 2023)            | CarDD (pixel-level)                          | Various (benchmark evaluation)      | Best mAP@50: ~0.71 across models             | Detection only; no downstream report generation             |
| YOLOv8 damage segmentation (IEEE, 2024) | Custom 4k images, 21 part + 8 damage classes | YOLOv8-seg                          | mAP@50: ~0.73 damage; ~0.81 parts            | No policy integration; severity not defined                 |
| HL-YOLO (MDPI, 2025)                    | Custom vehicle dataset                       | YOLO11 + heterogeneous convolutions | +3-4% mAP over YOLO11; absolute mAP@50 ~0.79 | Detection only; no NLP or policy component                  |
| Lewis et al. RAG (NeurIPS, 2020)        | NaturalQuestions, TriviaQA                   | DPR + BART                          | NQ EM: 44.5; TriviaQA EM: 56.8               | Text-only; hallucination risk without grounding             |



# Why Not Just Use a VLM?

Florence-2, Qwen2.5-VL, LLaVA, GPT-4V are compelling but the modular pipeline is a better fit

1

## Separation of Concerns & Debuggability

Each stage (detect, retrieve, generate) can be isolated and fixed independently. A monolithic VLM is a black box: a bad output gives no signal as to what went wrong, slowing iteration in a time-boxed project.

2

## Measurable, Ground-truth Comparable Detection

YOLO outputs bounding boxes and class labels, directly evaluable against VehiDE (mAP@50, F1 per class). VLM damage descriptions are free-form text, not comparable to bounding-box annotations without extra post-processing.

3

## Cost, Latency & Deployment

YOLO-nano/small runs on CPU and fits within Hugging Face Spaces limits. Large VLMs need paid APIs or dedicated GPU memory Spaces cannot reliably provide. Modular pipeline keeps each component lightweight and swappable.

# Gaps in Existing Solutions

## Gap 1

### Detection Without Structured Reporting

YOLO fine-tunes and CarDD benchmark models output bounding boxes and class indices not human-readable reports. No open-source system bridges detection output to a final claim assessment.

## Gap 2

### LLM Reports Without Policy Grounding

General-purpose LLMs (GPT-4, Gemini) hallucinate coverage entitlements, incorrect exclusions, and fabricated deductible values without access to the actual policy. RAG grounding is absent from all publicly demonstrated systems.

## Gap 3

### No Accessible Decision-Support Demo

Industry systems (Tractable, CCC) are closed and proprietary. There is no open, deployable vision-plus-language claim assessment tool. Our Hugging Face Spaces deployment addresses this gap directly.

# Our Contribution

Not a new model architecture three meaningful contributions:

## Open-Source End-to-End Pipeline

A publicly accessible system combining YOLO-based damage detection with policy-aware LLM report generation. Specifically designed for claim assessors, dramatically reducing time to first-draft assessment.

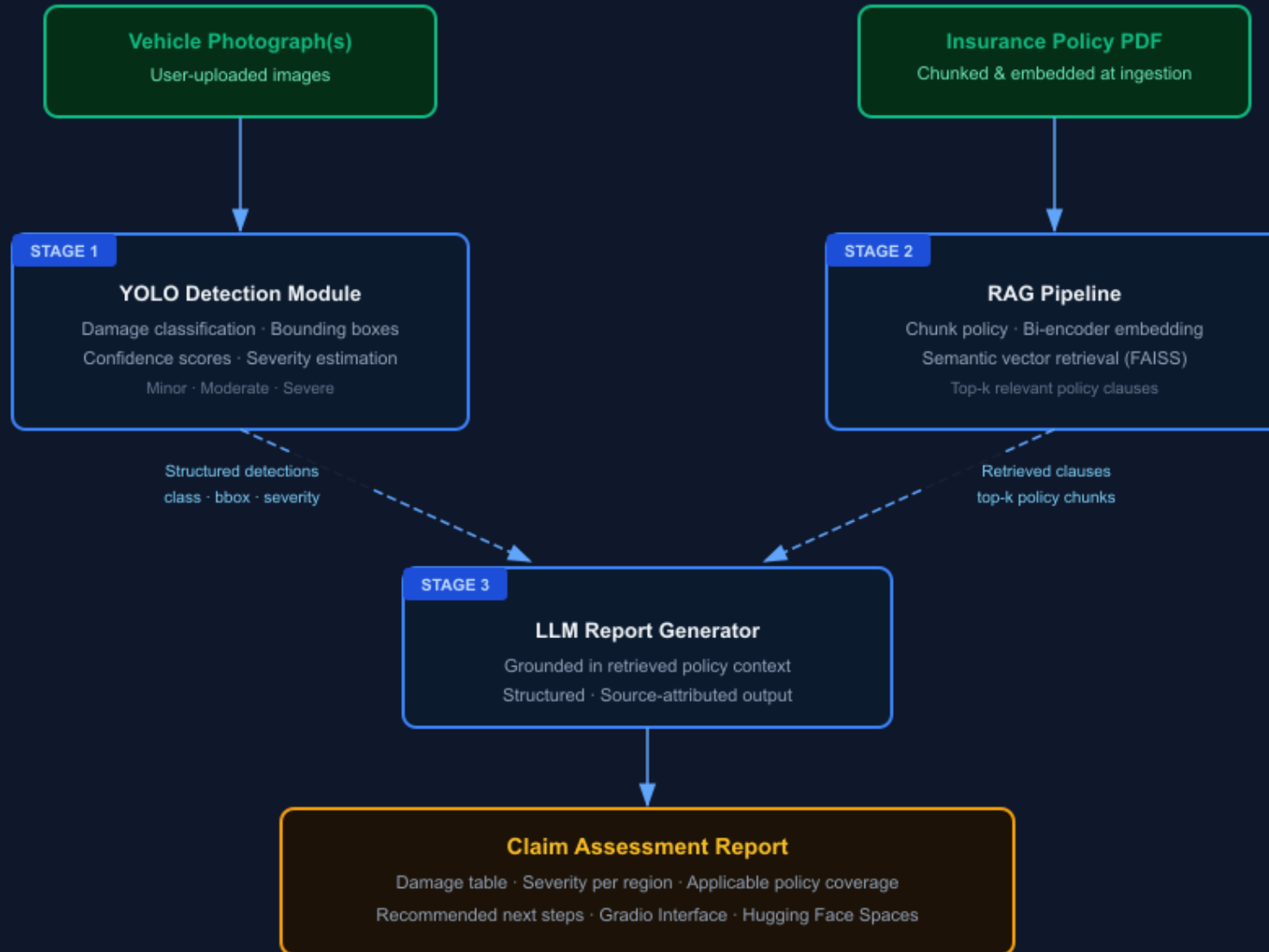
## Pipeline Integration

YOLO detections are converted to structured format, then fed to an LLM with RAG-retrieved policy clauses, enabling accurate, grounded reports. Integration of all three components is the core contribution.

## Accessible Deployment on Hugging Face Spaces

The first open, interactable demonstration of a vision-plus-language vehicle claim assessment tool — accessible to researchers and small insurers who cannot access proprietary industry systems.

# System Architecture



# Success Metrics

## Vision Model (YOLO)

| Metric                   | Target      |
|--------------------------|-------------|
| mAP@50                   | >= 0.70     |
| mAP@50-95                | >= 0.50     |
| Per-class F1 (6 classes) | >= 0.65     |
| Inference speed          | Report only |

### Target Justification

mAP@50 >= 0.70: production automotive standard, achievable on VehiDE with YOLO11.

Precision@3 >= 0.80: reflects Lewis et al. on comparable document retrieval.

Human eval >= 4.0: conservative on 5-point scale for genuine utility.

## RAG Pipeline

| Metric                | Target  |
|-----------------------|---------|
| Retrieval Precision@3 | >= 0.80 |
| Recall@3              | >= 0.75 |
| Faithfulness score    | >= 0.85 |

## End-to-End & Usability

| Metric                | Target      |
|-----------------------|-------------|
| Human eval / accuracy | Mean >= 4.0 |
| Human eval / clarity  | Mean >= 4.0 |
| Severity accuracy     | >= 0.75     |
| BERTScore F1          | >= 0.80     |
| Ablation delta        | Positive    |

# Evaluation Plan

## Vision

### Damage Detection

Standard object detection metrics on unseen test images:  
mAP@50, mAP@50-95, Precision, Recall, F1-score

## RAG

### Policy Retrieval

Retrieval Precision@3 measures whether correct policy clauses appear in top-3 results for each damage type.

## End-to-End

### Generated Reports

3 independent raters score 20 reports on Accuracy, Faithfulness, Clarity (1-5 scale).  
BERTScore F1 as objective signal.

## Human Evaluation Protocol

- **Sample:** 20 generated reports evaluated by 3 raters (2 team members + 1 external reviewer)
- **Dimensions:** Accuracy (reflects detected damage), Faithfulness (supported by retrieved clauses), Clarity (useful to assessors)
- **Rubric:** Score 5 requires all coverage recommendations traceable to retrieved clauses; score 1 indicates fabricated entitlements
- **Agreement:** Disagreements >1 point resolved by discussion and majority vote; inter-rater agreement reported via Cohen's kappa

# Dataset Plan

## Vision Datasets

| Dataset             | Size            | Role                                   | Justification for selection                                                                          |
|---------------------|-----------------|----------------------------------------|------------------------------------------------------------------------------------------------------|
| VehiDE              | 13,945 images   | Primary training & evaluation          | Largest publicly available annotated vehicle damage dataset                                          |
| CarDD               | Varies by split | Supplementary segmentation fine-tuning | Provides pixel-level masks unavailable in VehiDE                                                     |
| COCO Car Damage     | ~500 images     | Supplementary architecture comparison  | Allow direct integration with standard detection training pipelines and enable comparison baselines. |
| Car Damage Severity | ~2,300 images   | Severity classifier calibration        | Only publicly available dataset with human-assigned severity labels                                  |

## Policy Data to be created by Team

- 5 synthetic policy PDFs with collision, comprehensive, deductibles, exclusions, claim limits, third-party liability
- Coverage clauses mapped to all 6 damage classes for ground-truth RAG evaluation
- 50 synthetic incident descriptions paired with test images for end-to-end evaluation

# Risks & Mitigations

## Dataset Imbalance

Dents/scratches dominate rare classes. Mitigation: class-weighted loss, augmentation oversampling, per-class F1 monitoring.

## Severity Estimation

BBox-area proxy has failure modes. Mitigation: calibrate against Car Damage Severity dataset + human annotator consensus.

## Domain Shift

Training on studio images; real claims are mobile photos in variable conditions. Mitigation: brightness/contrast/perspective augmentation + realistic stress-test set.

## RAG Faithfulness (Synthetic Policies)

Synthetic docs may be simpler than real policies. Mitigation: varied phrasing, negations, and distractor clauses to stress-test retrieval.

## LLM Hallucination on Edge Cases

LLM may invent coverage when no clause is retrieved. Mitigation: explicit prompt instruction to state 'not covered' rather than infer from parametric knowledge.



# Ethical Considerations

Decision-support only: all final decisions remain with qualified human assessors.

## Fairness

### Bias in Decision Support

Underrepresented vehicle types or damage patterns may cause systematic underperformance. Mitigation: stratified error analysis across classes and vehicle types. All reports state they are AI-generated preliminary assessments.

## Privacy

### Data Protection

Vehicle photos and policy documents contain PII. No user data is retained beyond the session. PII masking via Microsoft Presidio SDK. Attribute shuffling for test records. Compliance with India's DPDP Act and GDPR.

## Transparency

### AI-Generated Reports

All reports include a disclaimer: preliminary AI-assisted assessment, not verified by a licensed assessor, not to be used as the sole basis for final claim decisions. System supports, not replaces, assessor judgment.

**System Role:** This is a first-pass decision-support tool. Every generated report is subject to mandatory human review before any claim decision. The AI does not approve, reject, or finalise claims.

# Thank You

## Questions & Discussion

**GitHub:** [HiveCase/Group-1-DS-and-AI-Lab-Project](#)

**Demo:** [Hugging Face Spaces](#) (to be deployed)

**Course:** DS & AI Lab · May 2026 · Group 1 · IIT Madras